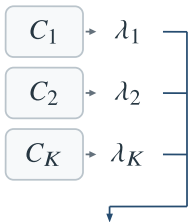


Penalty view

Costs in the objective



$$A_r - \sum_k \lambda_k A_{c_k}$$

SafeOT

One flow, all budgets

$$\langle C_k, F \rangle \leq b_k \quad \forall k$$



$\rightarrow$ observed $\Rightarrow$ target hard edge excluded

Entropic projection

$\lambda_k^* \rightarrow$ clip + EMA

observed-cost feedback

can price before violation

$$\bar{J}_k - B_k$$

$$\lambda_k = \bar{\lambda}_k^* + \beta_k$$

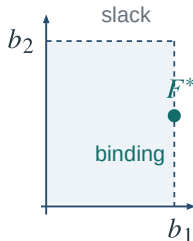
priced advantage

$\rightarrow$ **PPO/TRPO**

$\rightarrow \pi_{t+1}$

Constraint view

Budgets bound flow costs



$$\lambda_1^* > 0, \quad \lambda_2^* = 0$$

One price rule

exact settings

Fixed penalty

$$c = 0$$
$$\beta_k \text{ fixed}$$

Lagrangian

$$c = 0$$
$$\beta_k \text{ adaptive}$$

SafeOT

finite ε

conditional limits

LP

$$\varepsilon \rightarrow 0$$

Active-set

$$\varepsilon \rightarrow \infty$$

saturated clip